

B+ Tree Implementation and Performance Analysis

Comparison of Memory-only, Disk-only, and Hybrid B+ Trees on 10 Million Records

1. Objective

This project implements and compares three B+ Tree storage strategies: memory-only, disk-only, and hybrid memory-plus-disk. The benchmark measures insertion time, search time, range-query time, reported peak memory, disk usage, tree height, node count, and page I/O where applicable.

2. Dataset and Assumptions

Item	Value / assumption
Record count	10,000,000
CSV columns	key,payload
Key / payload length	8 / 64 characters
Page size	4,096 bytes (4 KiB), as configured
Configured B+ Tree order	50
Hybrid cache capacity	256 pages for the reported run
Storage structure	Memory-resident nodes; file-backed nodes; file-backed nodes plus LRU cache for hybrid

The dataset is not included in the repository because of its large size. The benchmark used a CSV with header key,payload. Actual node capacity depends on node layout and page serialization; configured order does not by itself guarantee every node occupies exactly one 4 KiB page.

3. Results on 10,000,000 Records

Metric	Memory-only	Disk-only	Hybrid
Insertion time (s)	16.460	501.178	324.017
Search time (μs)	1.133	69.433	14.900
Range-query time (ms)	20.634	746.445	447.387
Peak memory (MB)	1731.24	12.86	14.05
Disk usage (MiB)	0.00	1153.63	1153.63
Tree height	5	5	5
Total nodes	295,329	295,329	295,329
Page reads	0	47,987,602	23,227,276
Page writes	0	48,252,770	10,537,540
Cache hits	0	0	24,760,326
Cache misses	0	0	23,227,271
Reported checksum	75,497,688	75,497,688	75,497,688
Range rows	1,048,576	1,048,576	1,048,576

```
M /c/Users/Rahul Saxena/Downloads/bplus_csv_update/bplus_csv_update
```

```
Requested records: 10000000  
ORDER: 50, page size: 4096 bytes  
Range rows in [K00000000, K00FFFFFFF]: 1048576
```

Memory-only B+ Tree

```
=====
Records inserted : 10000000
Insertion time   : 16.460 seconds
Tree height      : 5
Total nodes      : 295329
Peak memory      : 1731.24 MB
Search time      : 1.133 us
Range time       : 20634.000 us
Checksum         : 75497688
Page reads       : 0
Page writes      : 0
Cache hits       : 0
Cache misses     : 0
Disk usage       : 0.00 MiB
Range rows in [K00000000, K00FFFFFFF]: 1048576
```

Disk-only B+ Tree

```
=====
Records inserted : 10000000
Insertion time   : 501.178 seconds
Tree height      : 5
Total nodes      : 295329
Peak memory      : 12.86 MB
Search time      : 69.433 us
Range time       : 746444.900 us
Checksum         : 75497688
Page reads       : 47987602
Page writes      : 48252770
Cache hits       : 0
Cache misses     : 0
Disk usage       : 1153.63 MiB
Range rows in [K00000000, K00FFFFFFF]: 1048576
```

Hybrid B+ Tree

```
=====
Records inserted : 10000000
Insertion time   : 324.017 seconds
Tree height      : 5
Total nodes      : 295329
Peak memory      : 14.05 MB
Search time      : 14.900 us
Range time       : 447386.900 us
Checksum         : 75497688
Page reads       : 23227276
Page writes      : 10537540
Cache hits       : 24760326
Cache misses     : 23227271
Disk usage       : 1153.63 MiB
```

Screenshot of stats on my machine

4. Performance Analysis

Memory-only B+ Tree

It achieved the lowest measured insertion, search, and range-query times. Its reported peak memory was approximately 1.69 GiB, which is the main trade-off.

Disk-only B+ Tree

It had the lowest reported peak memory (12.86 MB), but insertion took 501.178 seconds. Its high page-read and page-write counts reflect the overhead of disk-backed operations.

Hybrid B+ Tree

Compared with disk-only, insertion time was lower by approximately 35.3%, search was about 4.66 times faster, page reads were lower by approximately 51.6%, and page writes by approximately 78.2%. Range-query time was also lower in this 10M run (447.387 ms versus 746.445 ms).

5. Correctness and Limitations

All three modes reported the same aggregate checksum and range count. These are consistency signals, not complete proof of record-level correctness: the current checksum sums key and payload lengths rather than hashing their actual contents. Timings depend on the test machine, filesystem, compiler, and cache state. Peak-memory figures also depend on the platform and measurement method.

6. Conclusion

For this run, memory-only provided the best raw performance, disk-only had the lowest reported memory usage, and hybrid offered a compromise by improving disk-only insertion and search performance while retaining disk-backed storage. Comparing returned payloads against expected CSV values and independently validating range-query contents would strengthen the correctness evaluation.

7. Source Code and Reproduction

GitHub repository: <https://github.com/rahulSaxena28/bplus-tree>

Run from the project directory:

```
./bplus_benchmark.exe --csv data/dataset.csv --mode all --n 10000000 --cache-pages 256
```